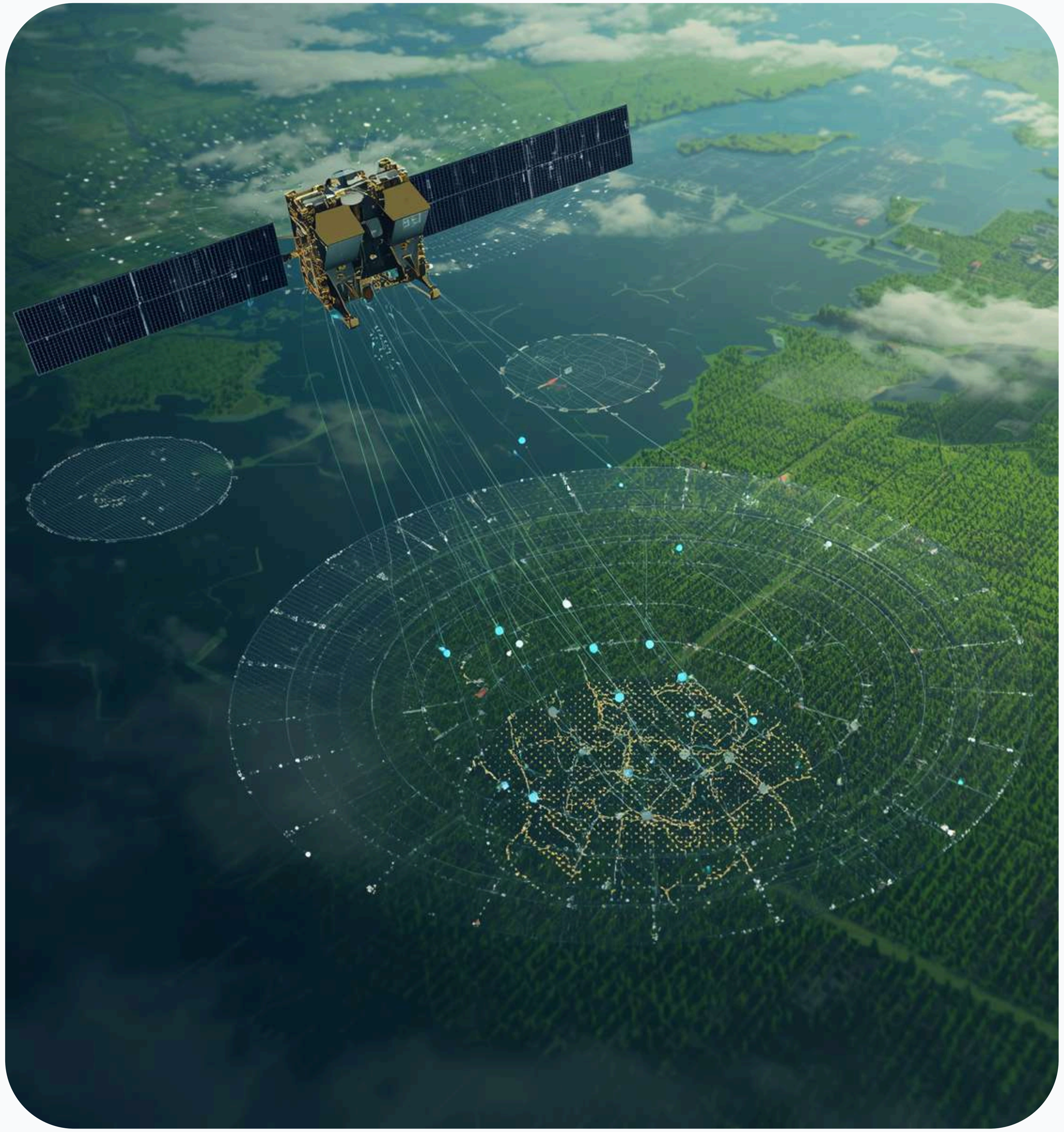
B.Tech Project 2026 IIT Delhi

Satellite-Based Crop Classification and Price Forecasting of Onion Using Machine Learning in indian agriculture



Application of ML on Onion Crop from Satellite Data

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Background of the Study

Onion cultivation faces significant challenges including years with bumper crops resulting in price crash, unpredictable yields, pest infestations, and climate variability. Traditional crop monitoring relies on manual field visits, which are time-consuming, labor-intensive, and limited in scale. Satellite remote sensing offers a transformative solution by providing continuous, large-scale observations of agricultural lands. This project explores how machine learning algorithms can analyze satellite imagery to monitor onion crop predict yields, and detect area under cultivation—addressing critical gaps in precision agriculture.

Research Questions

This study is guided by the following research questions:

01

How can satellite imagery drive economic and social benefits around Onion economy of India?

02

Which machine learning algorithms perform best for crop classification tasks?

03

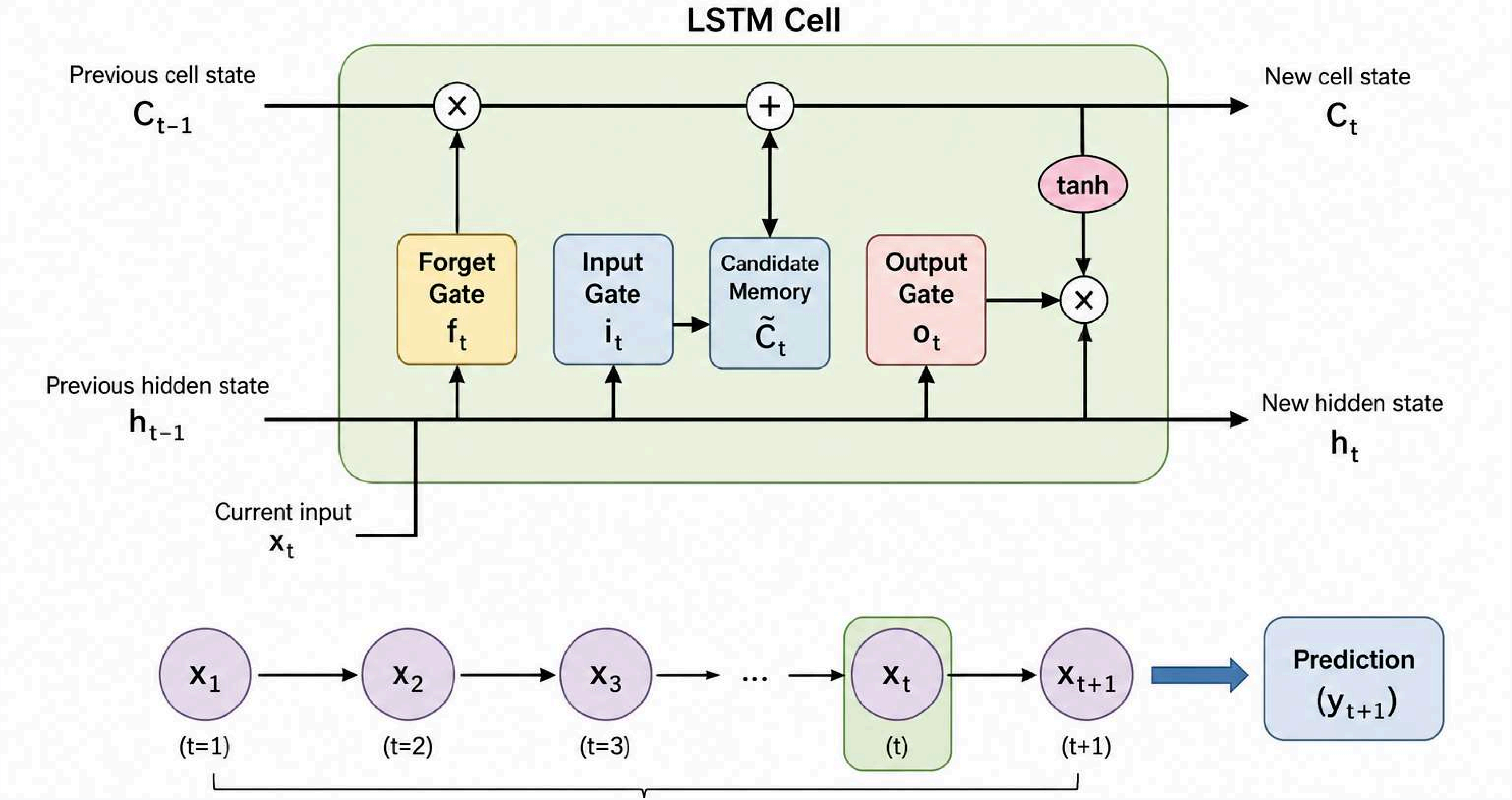
How accurate is yield prediction using remote sensing data analysis?

Onion price forecasting using lSTM

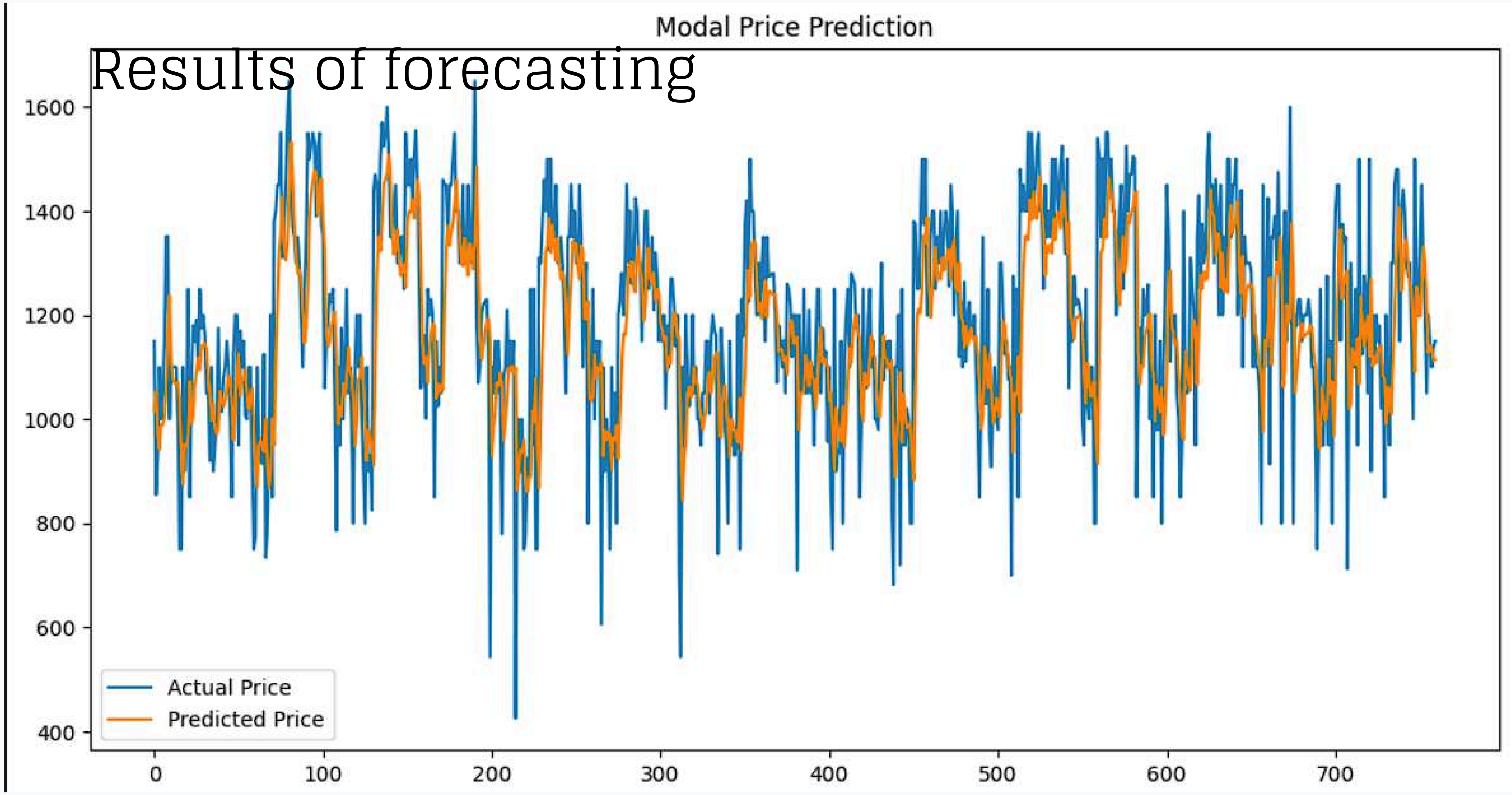
METHODOLOGY

- Dataset:
  - Nashik mandi data (2020–2025)
  - Weather data
  - 24,000+ records, 45 markets
- Preprocessing:
  - Missing values → Filling with mean values
  - Outliers → statistical filtering
  - Normalization:
- Model Architecture:
  - LSTM layers (temporal learning)
- Training:
  - Loss: RMSE.MAE,MAPE
  - Time based train-test split(2020-2024 train set,2025 test)

LSTM WORKING



Results of forecasting



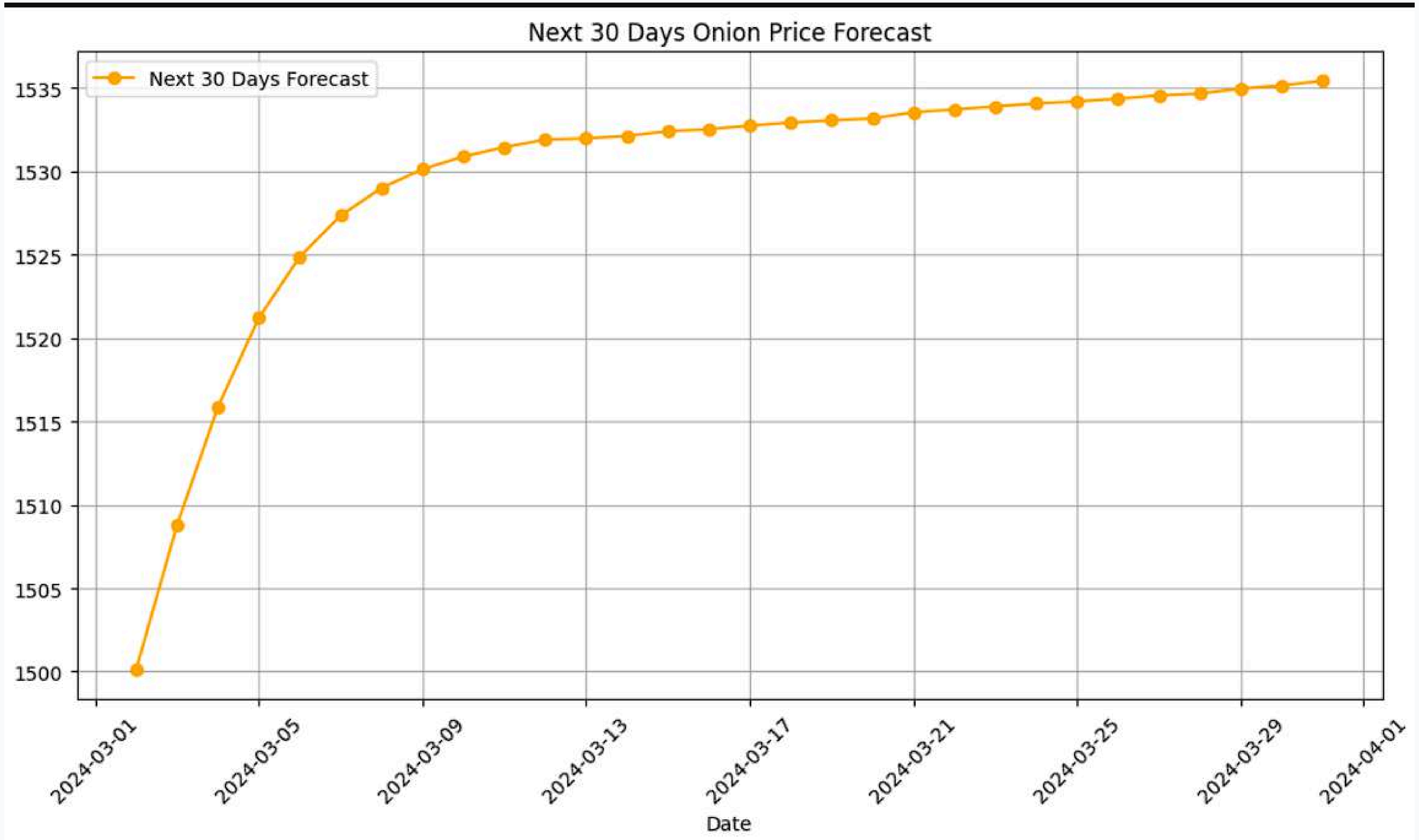
Results:

RMSE	164.45
MAE	124.22
MAPE	11.23%

Accuracy:approx 89%

R^2=0.34

Notice how LSTM captures trend and is unable to capture sudden spikes.This is due to the fact that Price volatility is primarily driven by production and supply dynamics, not just weather



Feature’s=[max_price,min_price,modal_price,Market,w2s,T2M,Rainfall,press,precpitation,date]

Engineered features=[lagged,rolled_mean]

Limitations of price forecasting

Observed Issues with LSTM Model

- **High variability across years**
- **Sudden price spikes difficult to capture**
- **Moderate predictive power (R2 = 0.34)**

Fundamental Limitation

- **The target feature “price” and weather inputs were insufficient(holds weak context about farmland)**
- **Does not directly capture:**
    - **Crop production levels**
    - **Area under cultivation**
    - **Crop health monitoring**

EDA insights

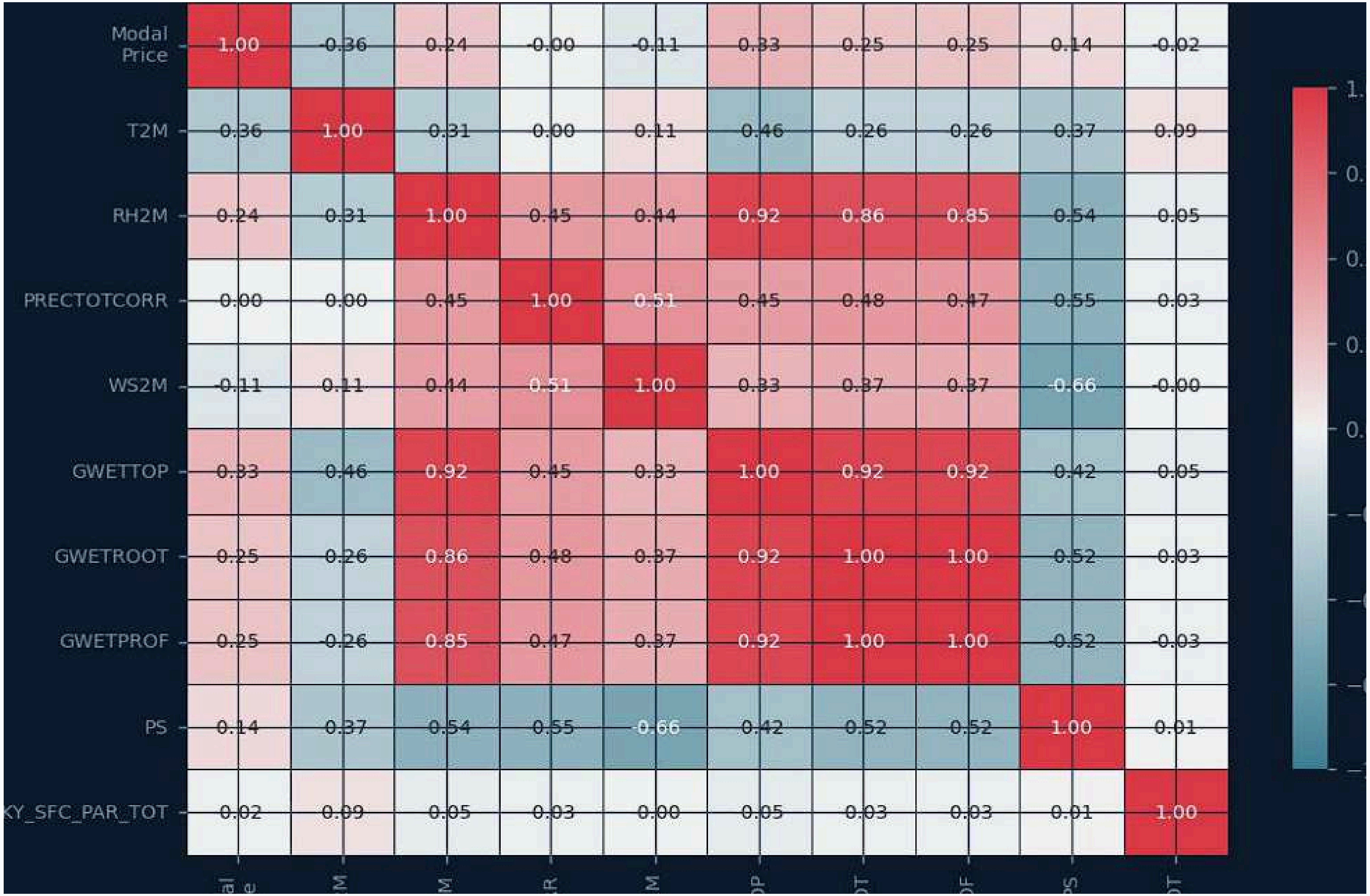
- **Price volatility driven at production level**
- **Weak correlation with weather alone**

Shift in Aprroach

- **Move from reactive prediction → proactive estimation**
- **Use satellite-based crop classification with bands features to capture spectral importance.**

NDVI=(B8-B4)/(B8+B4)

SAVI=((B8-B4)/(B8+B4+0.5))X1.5



The confusion matrix with weather inputs

Null hypothesis (incapability to predict spikes)

1. H0: The model is unable to accurately predict price spikes
2. H1: P(correct spike prediction)≤P0 (some baseline probability)

St=1 if spike occurs else 0

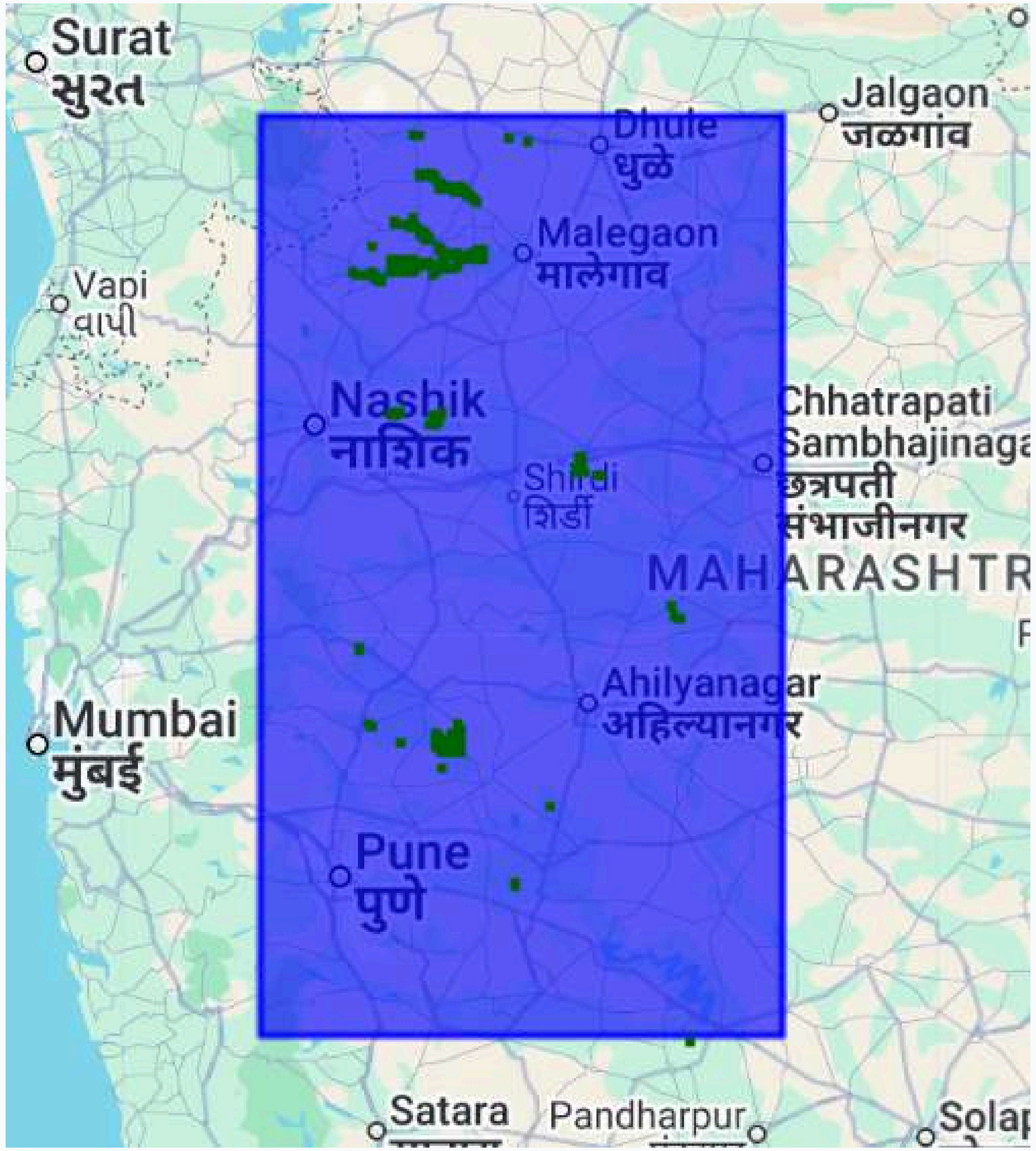
s^t=1 if model predicts spike else 0

- **Null hypothesis:**H0: E[(St−s^t)^2]≥E[(St−Stbaseline)^2]

  Stbaseline=1,if y{t}-y{t-1} > alpha else 0

- **Will pass at most alpha sized tests**

Classification

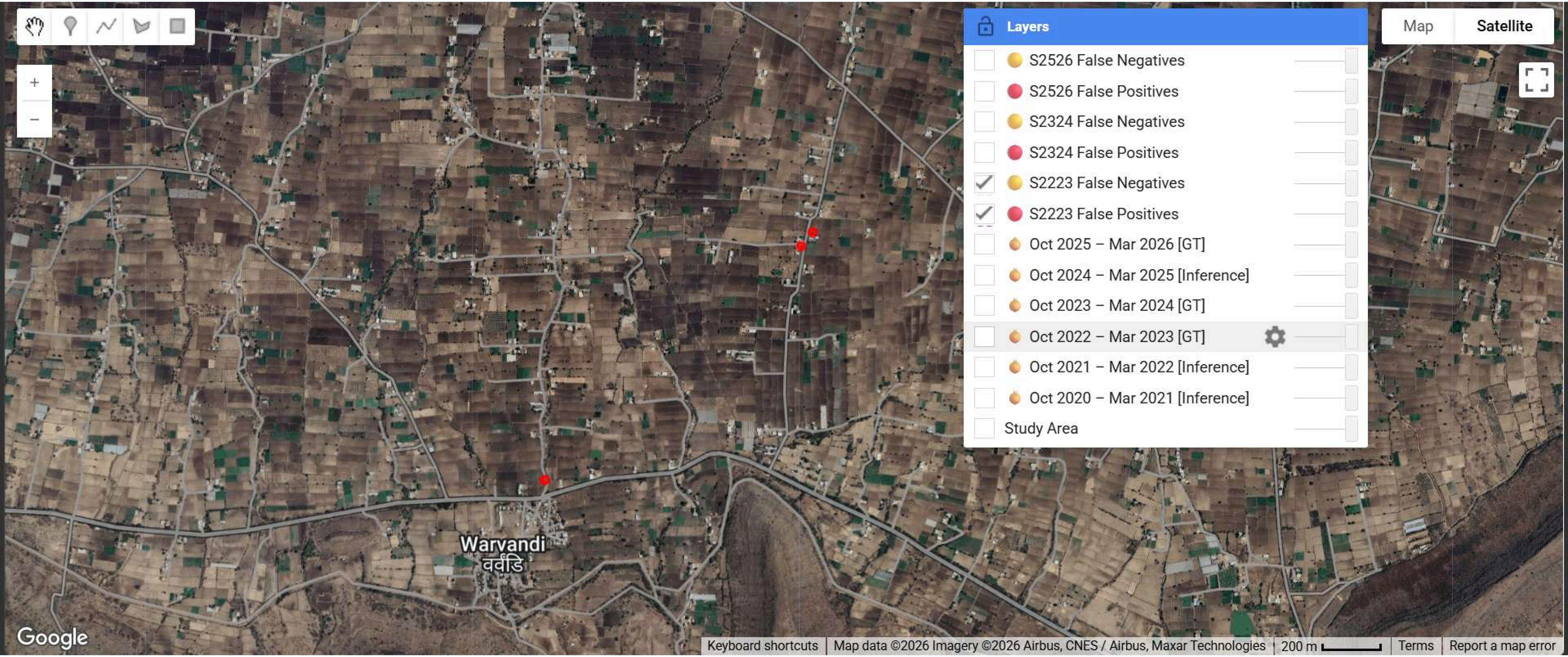


Feature engineering SELECTED_BANDS = [
'vh_peak', 'mois_d34', 'ndvi_season_max', 'mois_harvest_ratio',
'vh_season_mean', 'ndvi_m5', 'ndre_season_mean', 'latitude',
'vh_spatial_texture', 'ndti_m5', 'ndti_m4',
'chloro_gap_m4', 'chloro_gap_m3', 're_ratio_m4'];

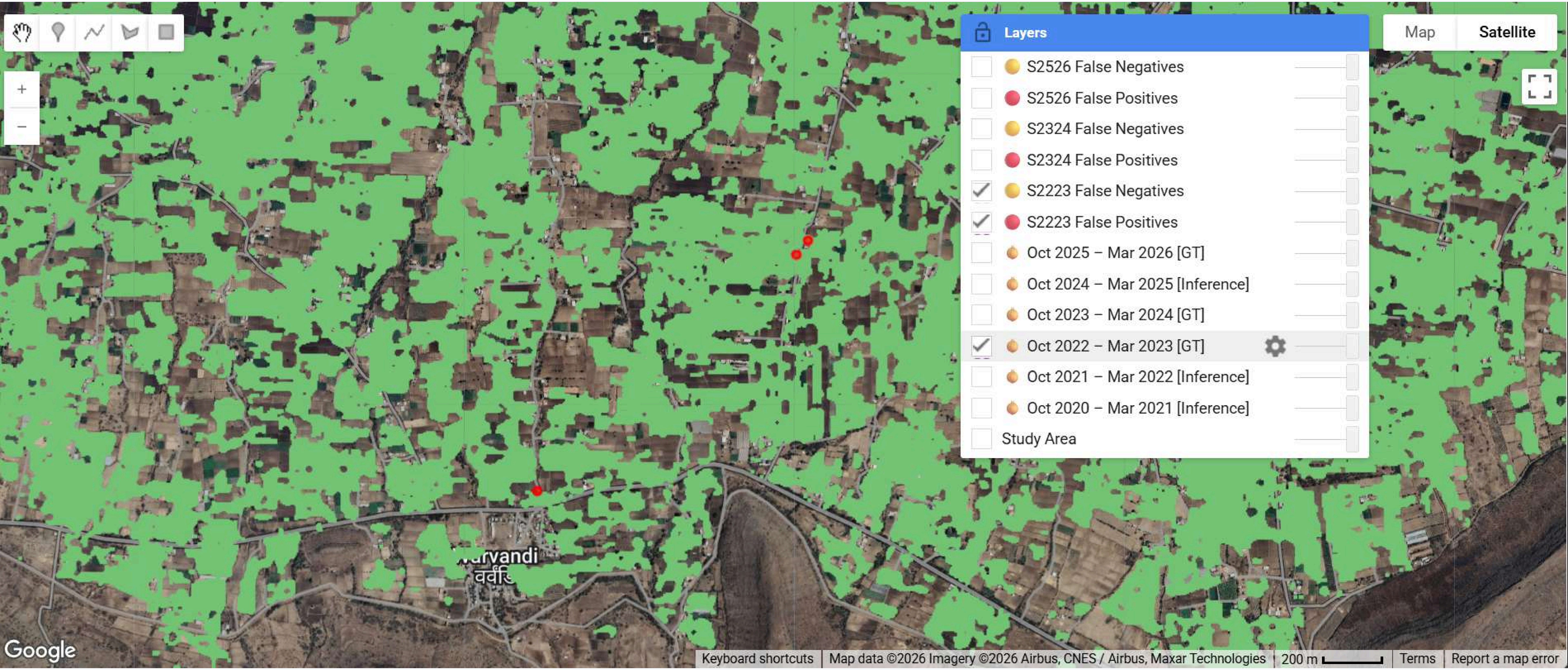
Test Season	Confusion Matrix	OA	κ
Oct 2022–Mar 2023	$\begin{bmatrix} 29 & 13 \\ 10 & 55 \end{bmatrix}$	0.7850	0.5435
Oct 2023–Mar 2024	$\begin{bmatrix} 39 & 24 \\ 12 & 46 \end{bmatrix}$	0.7025	0.4088
Oct 2025–Mar 2026	$\begin{bmatrix} 21 & 11 \\ 3 & 37 \end{bmatrix}$	0.8056	0.5962

Lower accuracy in 2024 can be attributed to GT data points being collected closer to roads or nearby fields rather than at the center of the field.

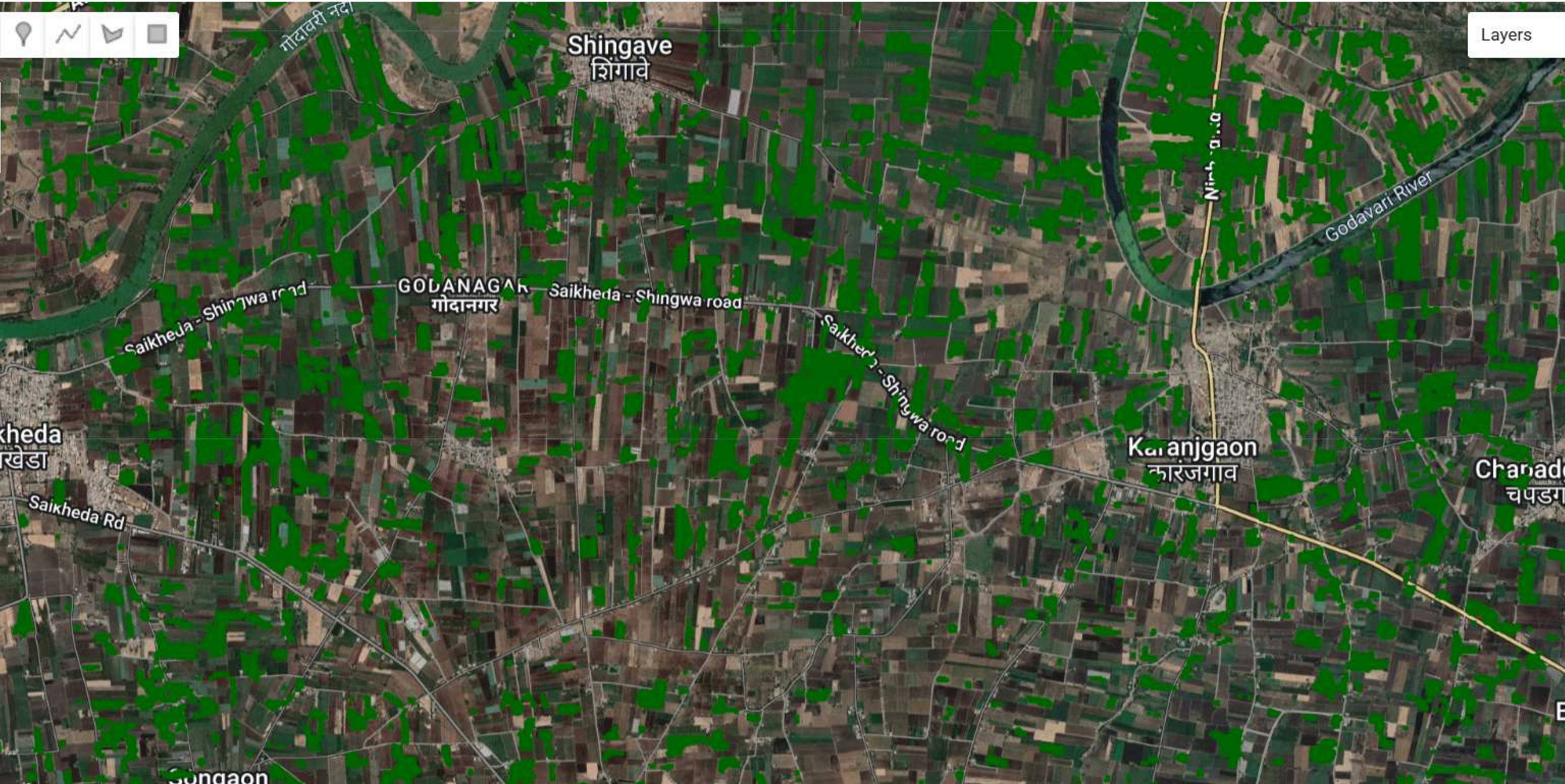
Classification



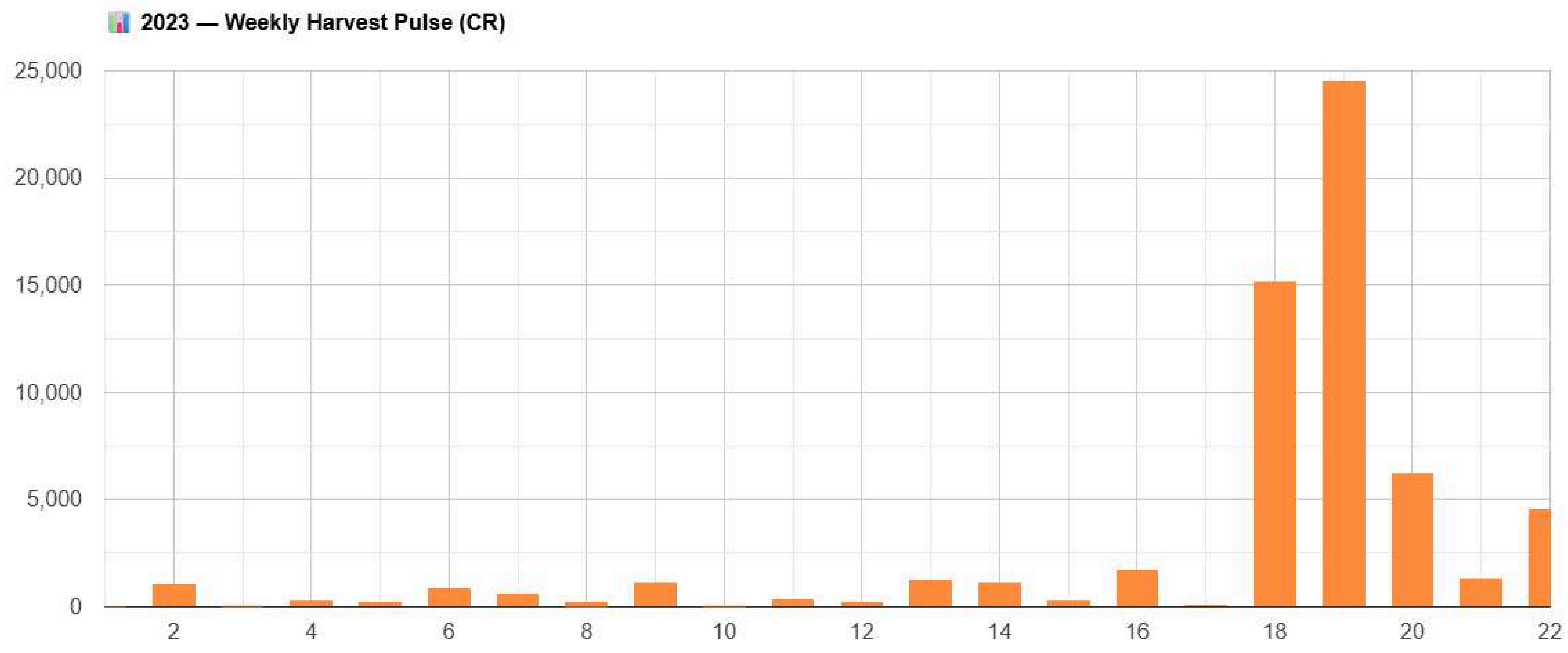
Classification

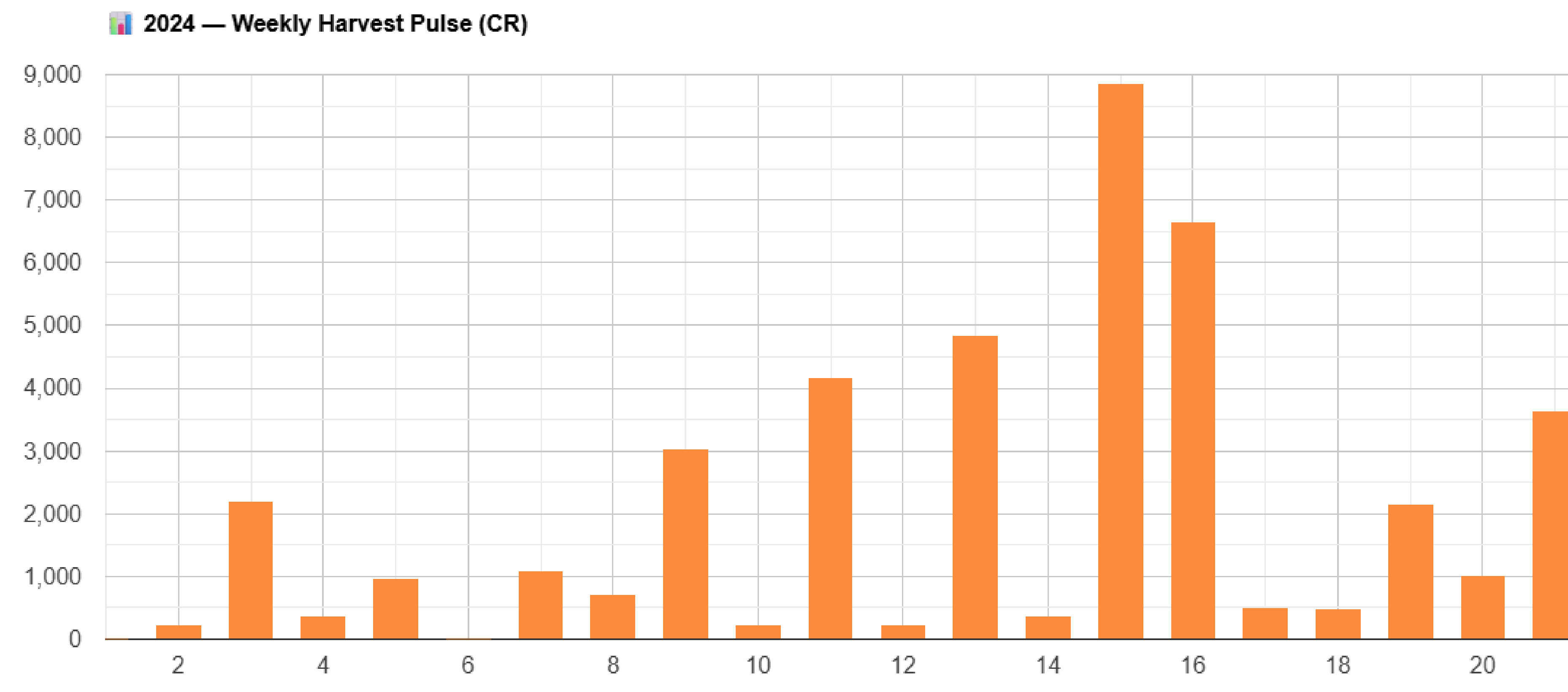


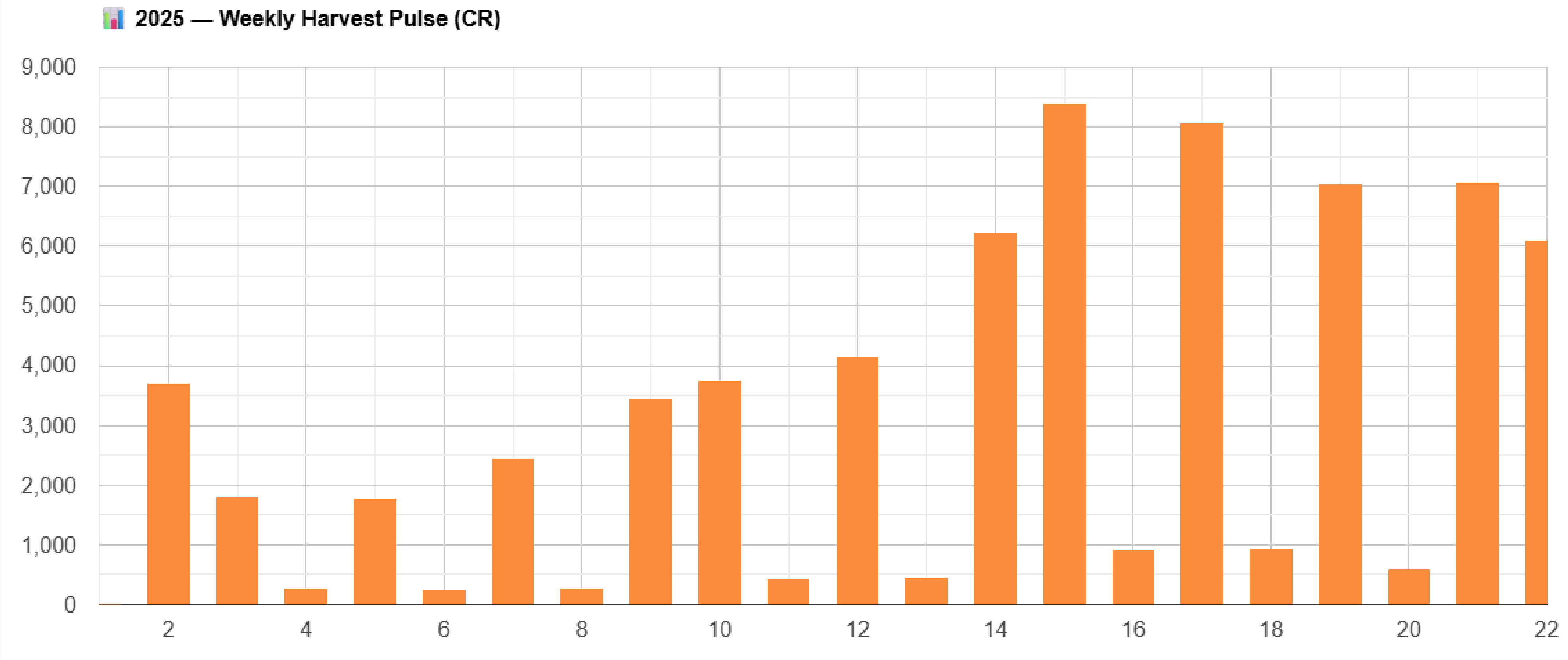
Classification

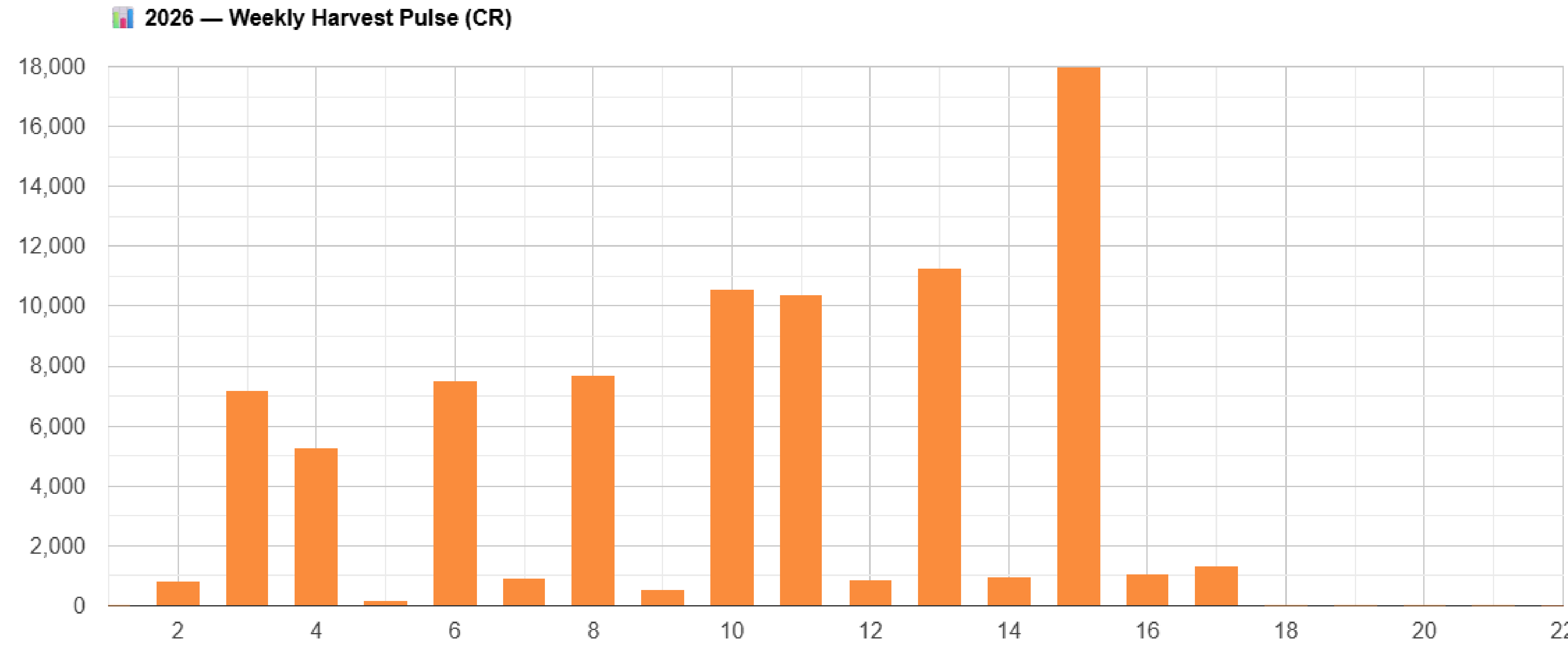


Harvest Progress





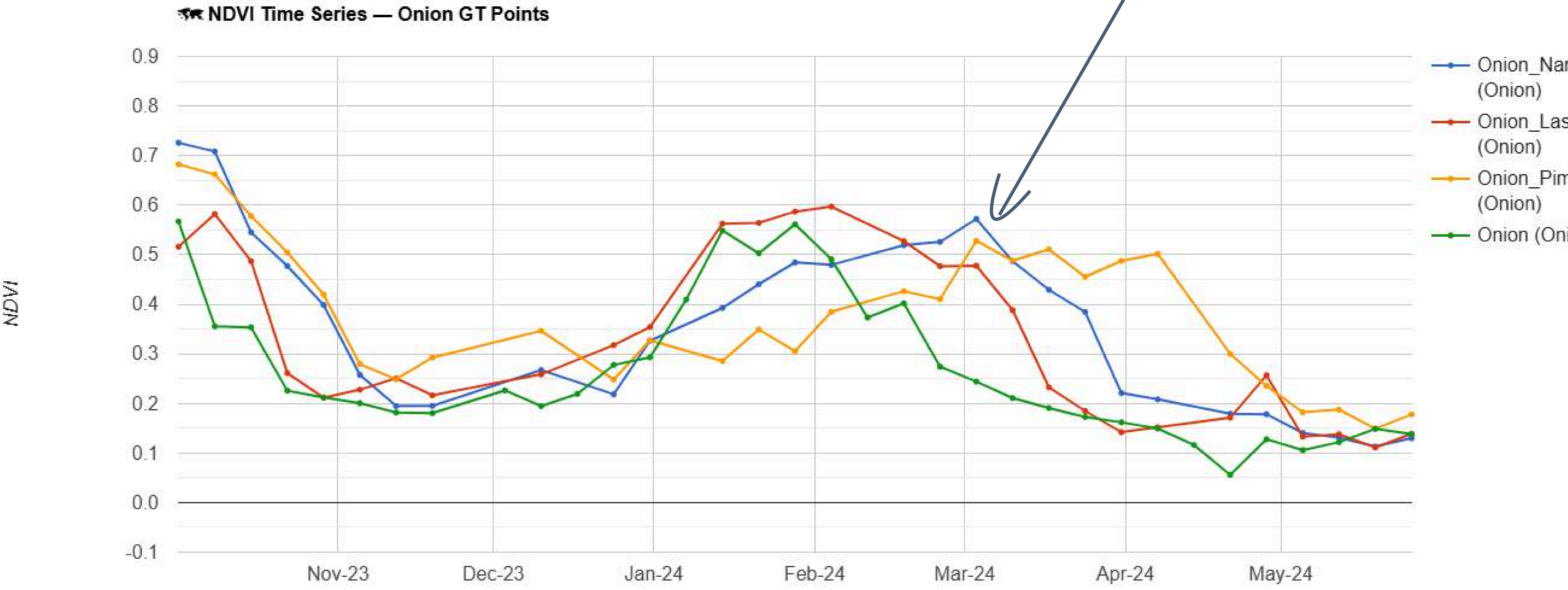




Bands Used for tracking harvest

No sharp dip to be sure the crop is harvested.

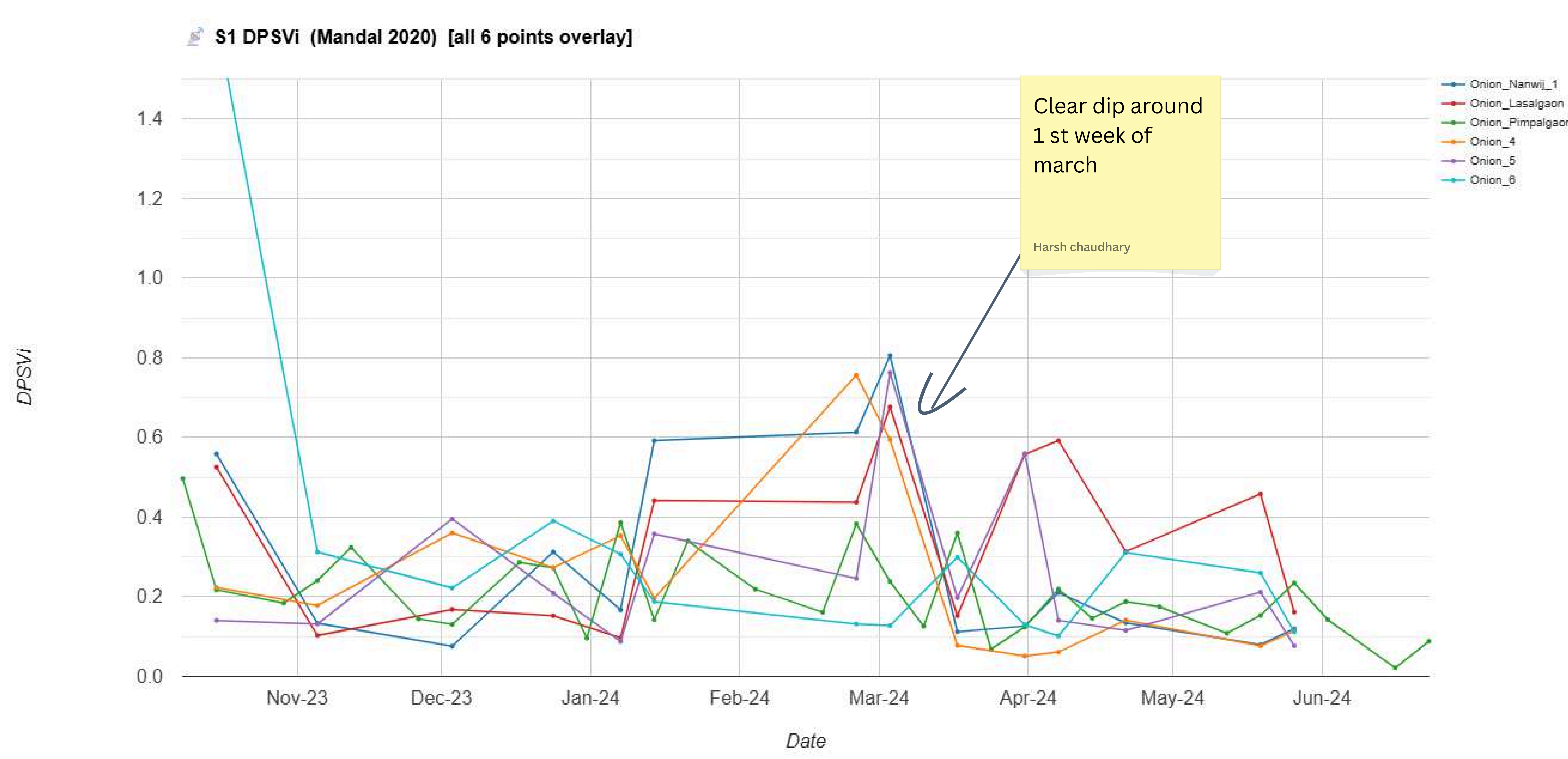
Harsh chaudhary



Bands Used for tracking harvest

Sentinel 1 can see through clouds but not 2.

DPSVi Drops sharply to ~0 at bare-soil transition.



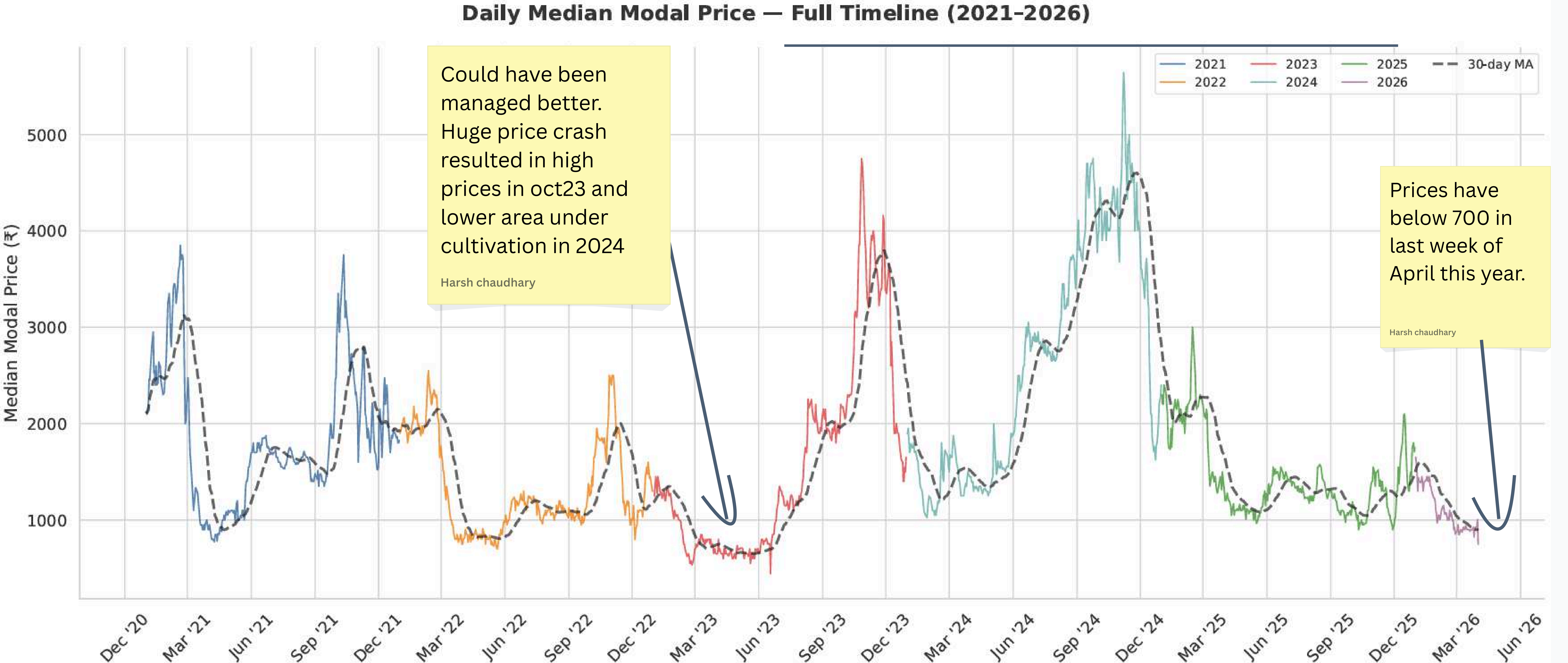
Mandi Prices

2023 62,255 Ha Area under cultivation

2024 42,860 Ha Area under cultivation

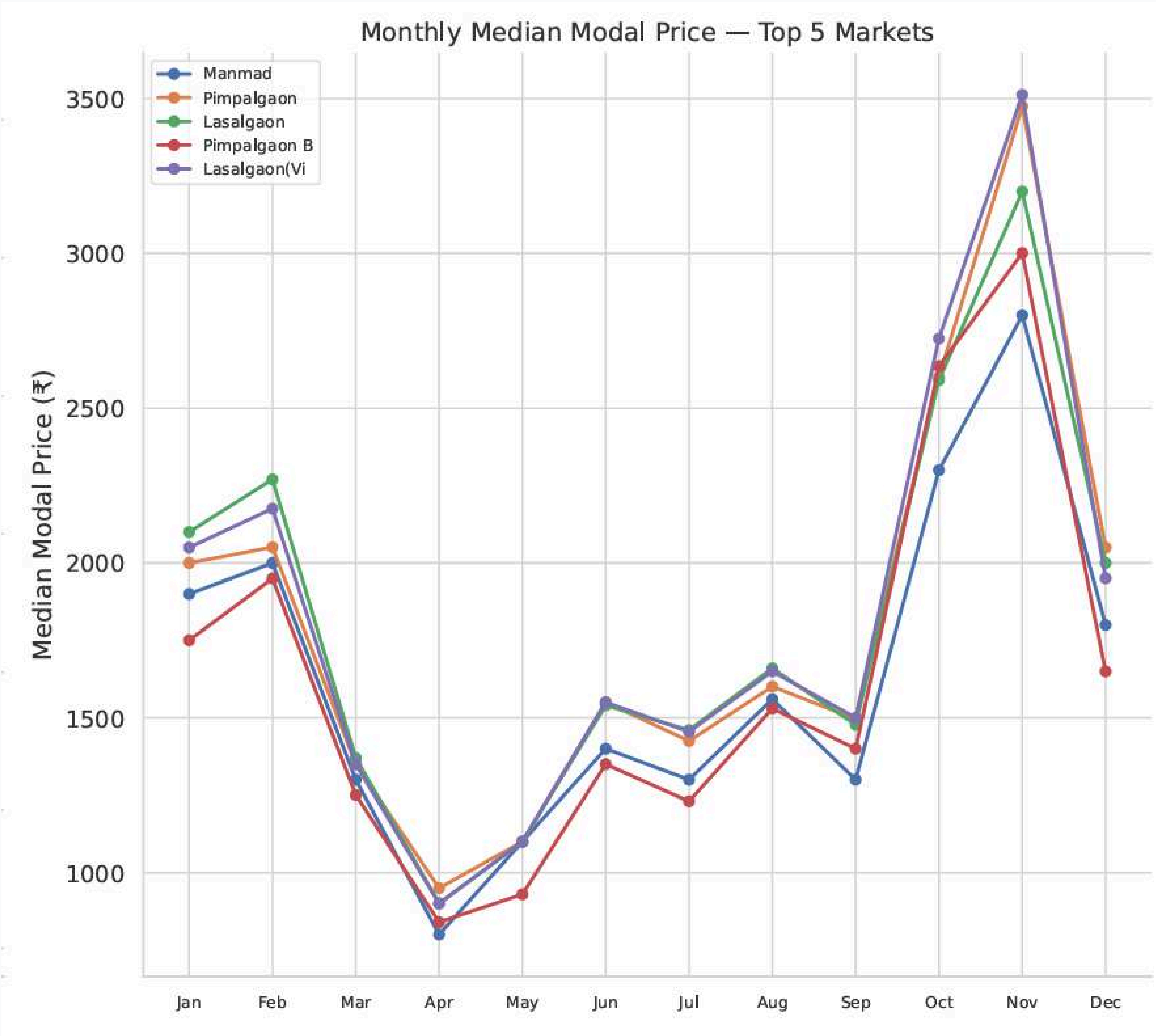
2025 69,105 Ha Area under cultivation

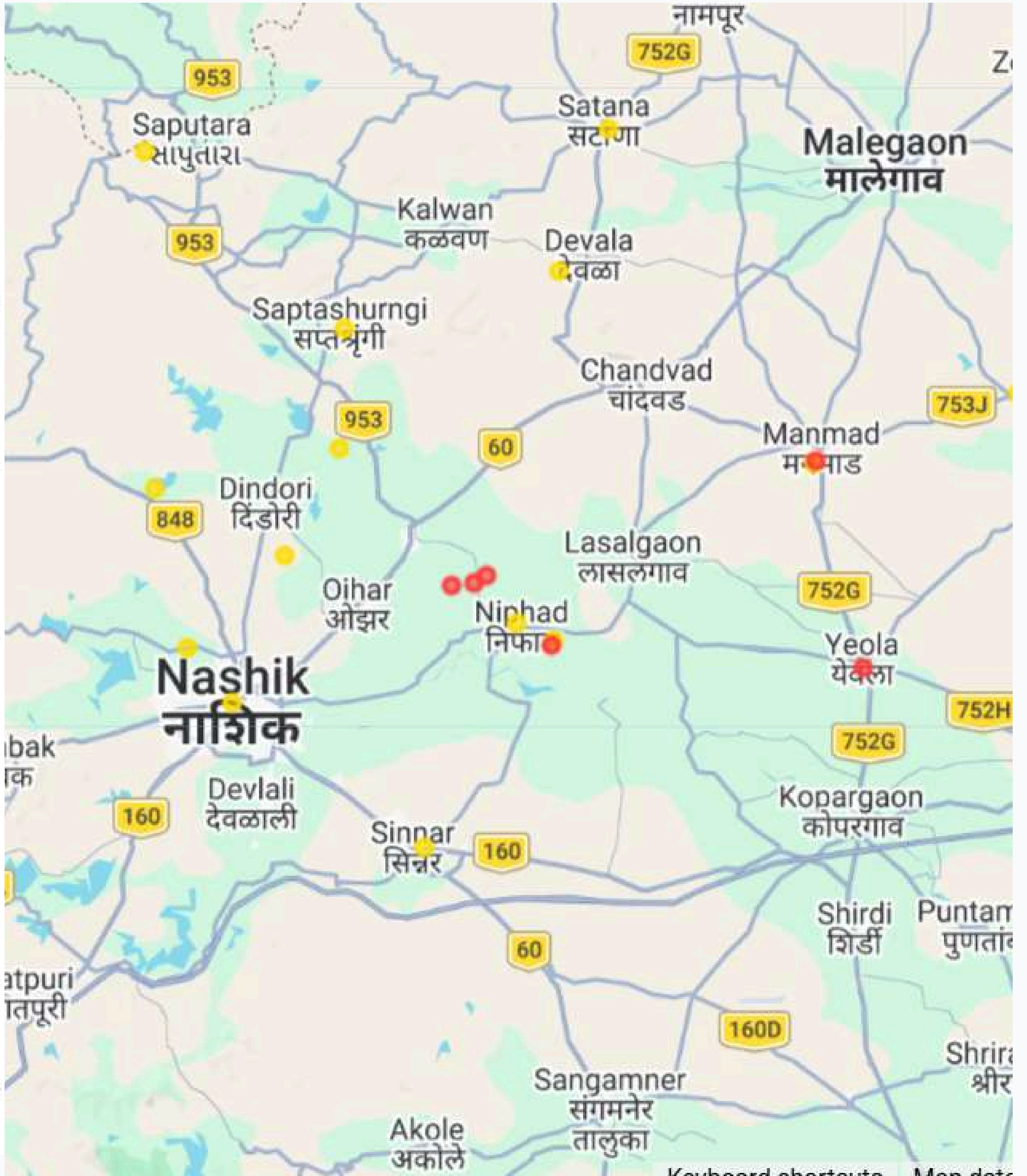
2026 83,540 Ha Area under cultivation



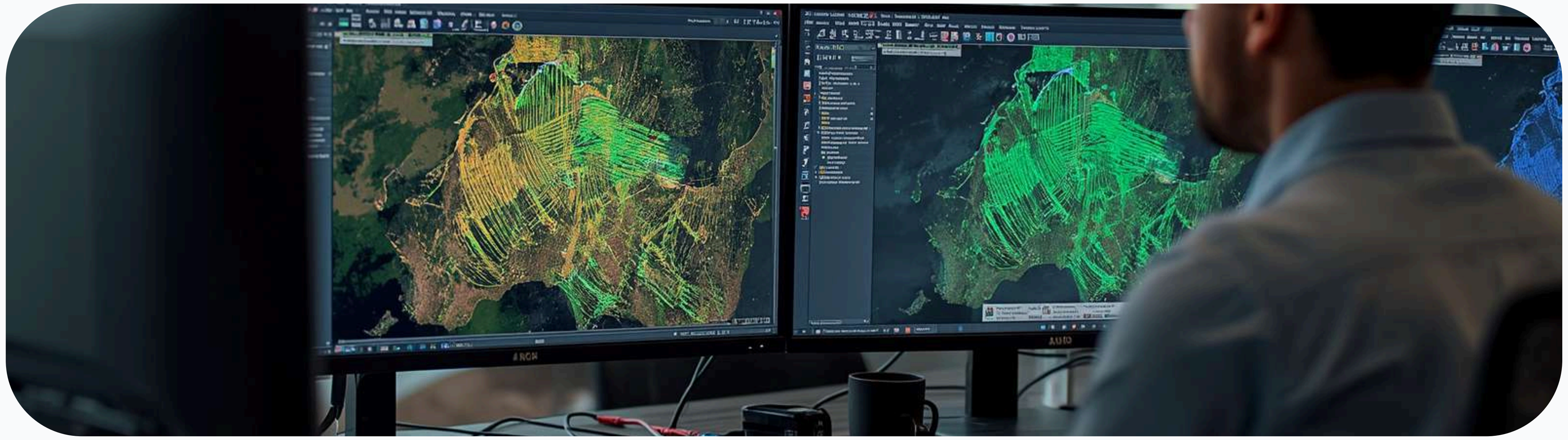
Mandi Prices

Across different geographic location Mandi





Key Findings



Our satellite-based ML analysis reveals key insights:

01 Significant variation of harvest pattern across geography and across Years

02 NDVI strongly correlates with onion growth stages

03 Anomaly in weather causes early harvest

These findings demonstrate the effectiveness of satellite-based ML for onion crop monitoring.

Practical Applications

Machine learning models trained on satellite data enable real-world agricultural solutions. Farmers can optimize water usage through precision irrigation, detect crop stress early to prevent pest outbreaks, and accurately forecast yields for better market decisions and supply chain management.

Application 1: Precision Agriculture & Targeted Irrigation

Application 2: Early Pest and Disease Detection

Application 3: Yield Estimation for Market Planning

Conclusions

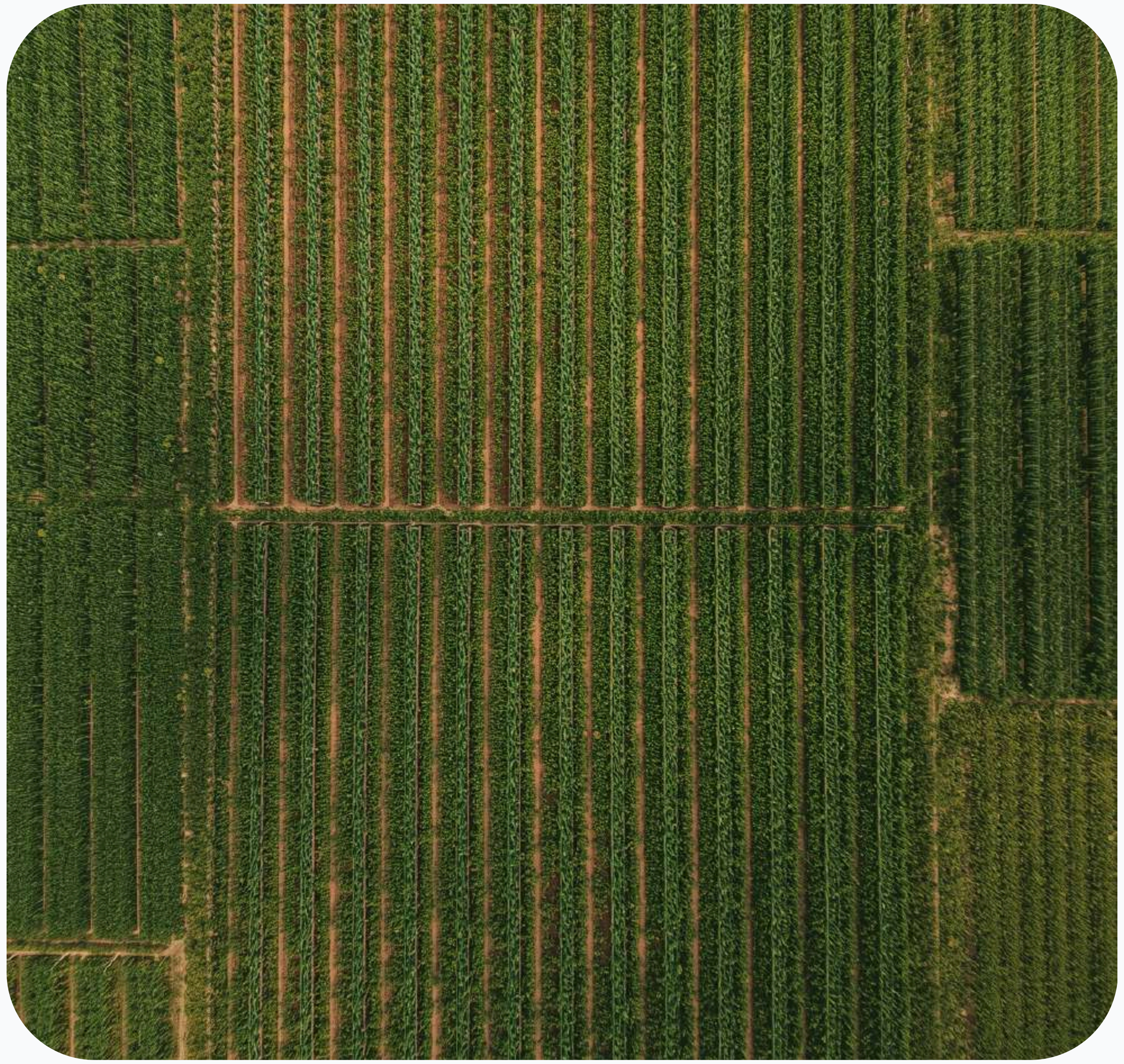
Key takeaways and future directions:

01 Onion crop classification using multi-temporal Sentinel data provides reliable estimation of area under cultivation

02 Finding yield using our classification model

03 Extension to other crops and larger geographic areas from Nashik

04 LSTM-based price forecasting captures seasonal trends but struggles with sudden spikes due to missing supply-side information



Future work

01 Use more advanced models like Transformers /or even hybrid models (LSTM + satellite features)

02 Improve ground truth quality and expand dataset across regions

03 Extend framework to other crops and larger geographic areas

Final remarks

Combining geospatial intelligence with machine learning can significantly improve both agricultural monitoring and market stability

Thank You!